

## SEMESTER S5

### REAL TIME OPERATING SYSTEMS

|                                            |                                        |                    |                |
|--------------------------------------------|----------------------------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEEVT 527</b>                       | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                                | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                                      | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PBEVT504,<br>PBECT404<br>(Course code) | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To develop skills in Real Time operating system to apply the same in real life applications.
2. Understand the concept of real time operating system along with its synchronization, communication and interrupt handling tools.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Contact Hours</b> |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Operating system: Types, Objectives and functions, Kernel, Process - States, Process Control Block, Operations on processes.                                                                                                                                                                                                                                                                                                                                   | <b>9</b>             |
| <b>2</b>          | Process Scheduling: FCFS, SJF, Priority, Round-Robin, Multilevel Queue and Multilevel Feedback Queue Scheduling. Thread: Structure. User and Kernel level threads, multi-threading models, multiprocessor scheduling.                                                                                                                                                                                                                                          | <b>9</b>             |
| <b>3</b>          | Real Time Operating Systems: Structure and characteristics of Real Time Systems, Task: Task states, Task synchronization -Semaphores- types, Inter Task communication mechanisms: message queues, pipes, event registers, signals, Exceptions and interrupt handling.                                                                                                                                                                                          | <b>9</b>             |
| <b>4</b>          | Task constraints, Task scheduling: Aperiodic task scheduling: EDD, EDF, LDF, EDF with precedence constraints. Periodic task scheduling: Rate monotonic and Deadline monotonic, Real time Kernel- Structure, State transition diagram, Kernel primitives.<br>Features of Free RTOS and Linux, Case study of (Kernel design, threads and task scheduling) RTOS: MicroC/OS-II,<br>RTOS control system used in real life applications - in adaptive cruise Control | <b>11</b>            |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal<br>Examination- 2<br>(Written) | Total |
|------------|-----------------------------|----------------------------------------|-----------------------------------------|-------|
| 5          | 15                          | 10                                     | 10                                      | 40    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                                                        | Part B                                                                                                                                                                                                                                                                                                       | Total     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                                                                    | Bloom's<br>Knowledge<br>Level (KL) |
|----------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>CO1</b>     | Summarize the functions and structure of general-purpose operating systems.                                        | <b>K2</b>                          |
| <b>CO2</b>     | Use different scheduling algorithms on processes and threads.                                                      | <b>K3</b>                          |
| <b>CO3</b>     | Interpret a real time operating system along with its synchronization, communication and interrupt handling tools. | <b>K2</b>                          |
| <b>CO4</b>     | Illustrate task constraints and analyse the different scheduling algorithms on tasks.                              | <b>K4</b>                          |
| <b>CO5</b>     | Illustrate the applications of real time operating systems.                                                        | <b>K3</b>                          |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 2   | 2   | 3   |     | 1   |     |     |     |     |      |      | 2    |
| <b>CO2</b> | 2   | 3   | 2   |     |     |     |     |     |     |      | 2    | 2    |
| <b>CO3</b> | 3   | 3   | 2   |     | 2   |     |     |     |     |      | 3    | 2    |
| <b>CO4</b> | 3   | 2   | 3   |     |     |     |     |     |     |      | 2    | 2    |
| <b>CO5</b> | 3   | 3   | 2   |     | 1   |     |     |     |     |      | 2    | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                                                                        |                             |                              |                               |
|-------------------|----------------------------------------------------------------------------------------|-----------------------------|------------------------------|-------------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                                                               | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b>       |
| 1                 | Operating System Principles                                                            | Abraham Silberschatz        | Wiley India                  | 7th edition, 2011             |
| 2                 | Operating systems- Internals and design principles                                     | William Stallings           | Prentice Hall                | 7th edition, 2011             |
| 3                 | Real-Time Concepts for Embedded Systems                                                | Qing Li                     | CMP Books                    | 2 <sup>nd</sup> Edition, 2013 |
| 4                 | HARD REAL-TIME COMPUTING SYSTEMS<br>Predictable Scheduling Algorithms and Applications | Giorgio C. Buttazzo         | Kluwer Academic Publishers   | 1 <sup>st</sup> Edition, 1997 |

| <b>Reference Books</b> |                                        |                             |                              |                               |
|------------------------|----------------------------------------|-----------------------------|------------------------------|-------------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>               | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b>       |
| 1                      | Modern Operating Systems               | Tanenbaum                   | Pearson Edition              | 3 <sup>rd</sup> Edition, 2007 |
| 2                      | Micro C/OS-II, The Real Time Kernel    | Jean J Lambrosse            | CMP Books                    | 2 <sup>nd</sup> Edition, 2011 |
| 3                      | Real-Time Systems: Theory and Practice | Rajib Mall                  | Pearson                      | 1 <sup>st</sup> Edition, 2008 |
| 4                      | An Embedded Software Primer            | David E Simon               | Pearson                      | 2 <sup>nd</sup> Edition, 2012 |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                              |
| 1                              | <a href="https://archive.nptel.ac.in/courses/106/105/106105172/">https://archive.nptel.ac.in/courses/106/105/106105172/</a> - Week 1 |
| 2                              | <a href="https://archive.nptel.ac.in/courses/106/105/106105172/">https://archive.nptel.ac.in/courses/106/105/106105172/</a> - Week 2 |
| 3                              | <a href="https://archive.nptel.ac.in/courses/106/105/106105172/">https://archive.nptel.ac.in/courses/106/105/106105172/</a> - Week 3 |
| 4                              | <a href="https://archive.nptel.ac.in/courses/106/105/106105172/">https://archive.nptel.ac.in/courses/106/105/106105172/</a> - Week 4 |